THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The largest possible solid sphere is carved out of a cube of edge 21 cm. The volume of the material wasted is: 1

(a) 4410 cm³ **(b)** 6835.5 cm³ **(c)** 4851 cm³ **(d)** 9261 cm³

2. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70, 70–80 and frequencies 3, 4, p, 4, 15, 10, 19, 22 respectively is 49. The value of p is: 1

(a) 22 **(b)** 25 **(c)** 23 **(d)** 28

3. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum of the numbers is 8 is: 1

(a) $\frac{1}{9}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{31}{36}$ **(d)** $\frac{5}{36}$

4. A solid sphere of radius 10.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). The volume of the space in the cylinder not occupied by the sphere is: 1

(a) 2425.5 cm³ **(b)** 12127.5 cm³ **(c)** 7276.5 cm³ **(d)** 4851 cm³

5. The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270 and frequencies 12, 8, x, 25, y, 18, 19 respectively, and total frequency 100, is 238.4. The value of x is: 1

(a) 10 **(b)** 8 **(c)** 11 **(d)** 9

6. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum is less than 5 is: 1

(a) $\frac{5}{36}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{7}{36}$ **(d)** $\frac{1}{6}$

7. A solid cylinder of radius 6 cm and height 5 cm is melted and recast into a cone of the same base radius. The height of the cone is: 1

(a) 30 cm **(b)** 15 cm **(c)** 10 cm **(d)** 5 cm

8. For the data with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 with frequencies 23, 4, 25, 6, 3, 19, 20 respectively, The mean of the data is: 1

(a) 269.8 **(b)** 270.8 **(c)** 268.8 **(d)** 270

9. A bag contains 15 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. The value of x is: 1

(a) 60 **(b)** 45 **(c)** 44 **(d)** 3

10. A solid cone and a solid hemisphere have equal base radii 7 cm and equal volumes. The height of the cone is: 1

(a) 10.5 cm **(b)** 21 cm **(c)** 14 cm **(d)** 7 cm

11. For the data with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 with frequencies 3, 13, 11, 17, 15, 24, 17 respectively, The median of the data is: 1

(a) 120 **(b)** 103.6 **(c)** 108 **(d)** 100

12. A bag contains 7 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{7}$. The value of x is: 1

(a) 4 **(b)** 10 **(c)** 6 **(d)** 5

13. The radii of two solid spheres are in the ratio 2 : 3. The ratio of their surface areas is: 1

(a) 2 : 3 **(b)** 8 : 9 **(c)** 4 : 9 **(d)** 8 : 27

14. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 17, 21, 24, 17, 6, 4, 11 respectively, The mode of the data is: 1

(a) 63 **(b)** 68 **(c)** 70 **(d)** 60

15. A bag contains 11 red and 5 blue balls. 9 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{4}{5}$ **(c)** $\frac{19}{25}$ **(d)** $\frac{21}{25}$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A solid sphere of radius 6 cm is completely dropped into water in a cylindrical vessel of base radius 8 cm, and no water spills. Find the rise in the water level. 2

17. The mean of a distribution with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 and frequencies 3, 24, p, 10, 15, 14, 17 respectively is 274. Find the value of p. 2

18. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is a multiple of 4. 2

19. A hemispherical bowl has outer radius 7.5 cm and inner radius 4.5 cm. Find the volume of material used to make the bowl. 2

20. The marks obtained by students in a test are given below. 2

Marks	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Number of students	3	4	23	4	15	10	19	22

Find the mode.

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A solid metallic cylinder of base radius 9 cm and height 32 cm is melted and recast into solid spheres of radius 6 cm. Find the number of spheres obtained. 3

22. The marks obtained by students in a test are given below. 3

Marks	40–50	50–60	60–70	70–80	80–90	90–100	100–110	110–120
Number of students	11	19	8	20	23	2	13	4

Find the mean and the mode (in this order).

23. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is 9. 3

24. A solid metallic cone of base radius 10 cm and height 5 cm is melted and recast into a solid sphere. Find the radius of the sphere. 3

OR

The marks obtained by students in a test are given below. 3

Marks	40–50	50–60	60–70	70–80	80–90	90–100	100–110
Number of students	17	21	24	17	6	4	11

Find the mean.

25. A bag contains 9 red and 7 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{2}{3}$. Find the value of x. 3

OR

A solid cylinder of base radius 6 cm and height 20 cm is melted and recast into identical solid cones, each of base radius 3 cm and height 8 cm. Find the number of cones obtained. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A toy is a cone of base radius 49 cm and height 168 cm mounted on a hemisphere of the same radius (the flat faces coincide). Find the total surface area of the toy. 5

OR

The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270 and frequencies 12, 8, x, 25, y, 18, 19 respectively, and total frequency 100, is 238.4. Find the value of x. 5

27. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the product of the numbers is 12,
(ii) the product is a perfect square,
(iii) the first number is greater than the second.

OR

A solid sphere of radius 31.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). Find the volume of the space in the cylinder not occupied by the sphere. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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